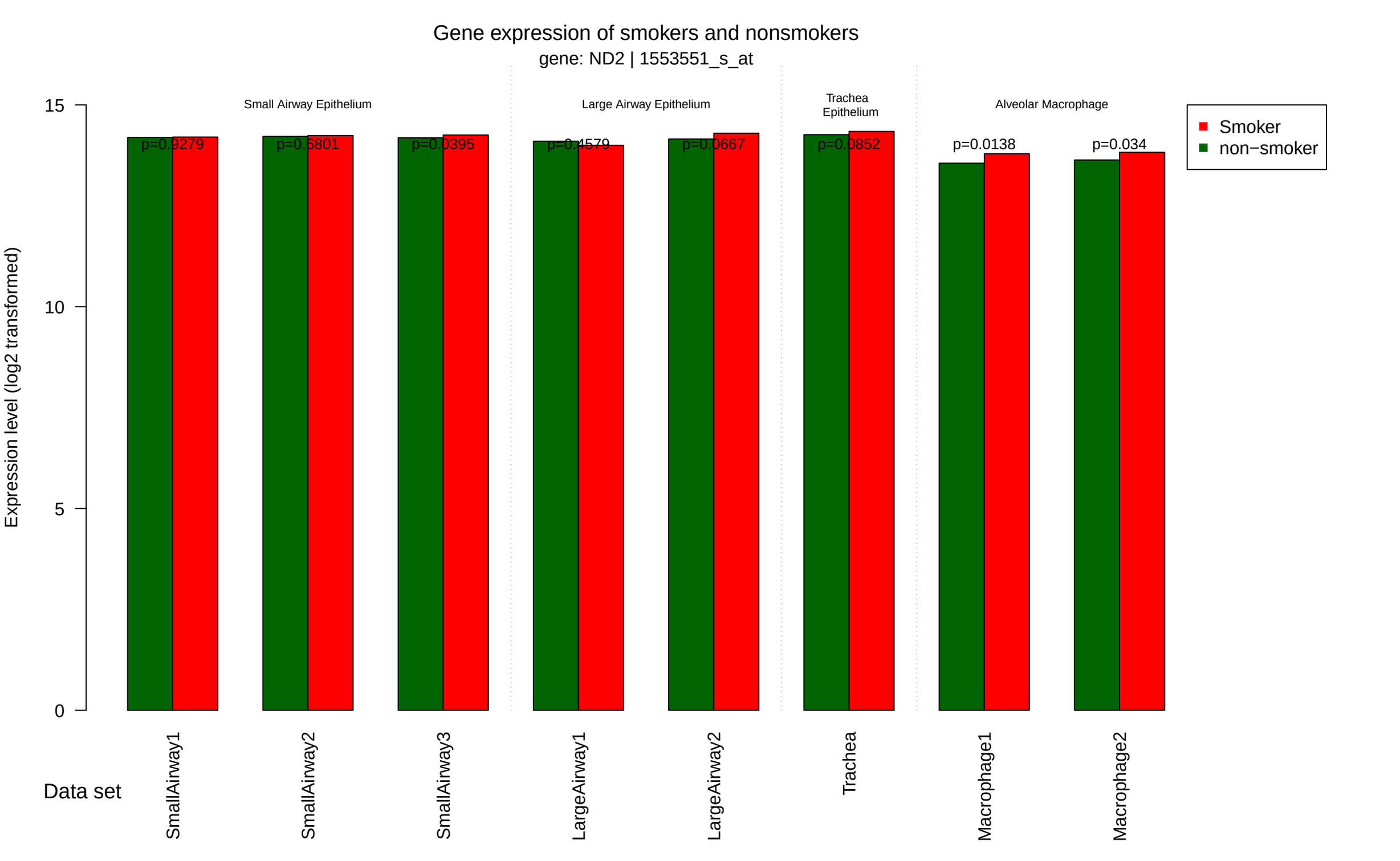
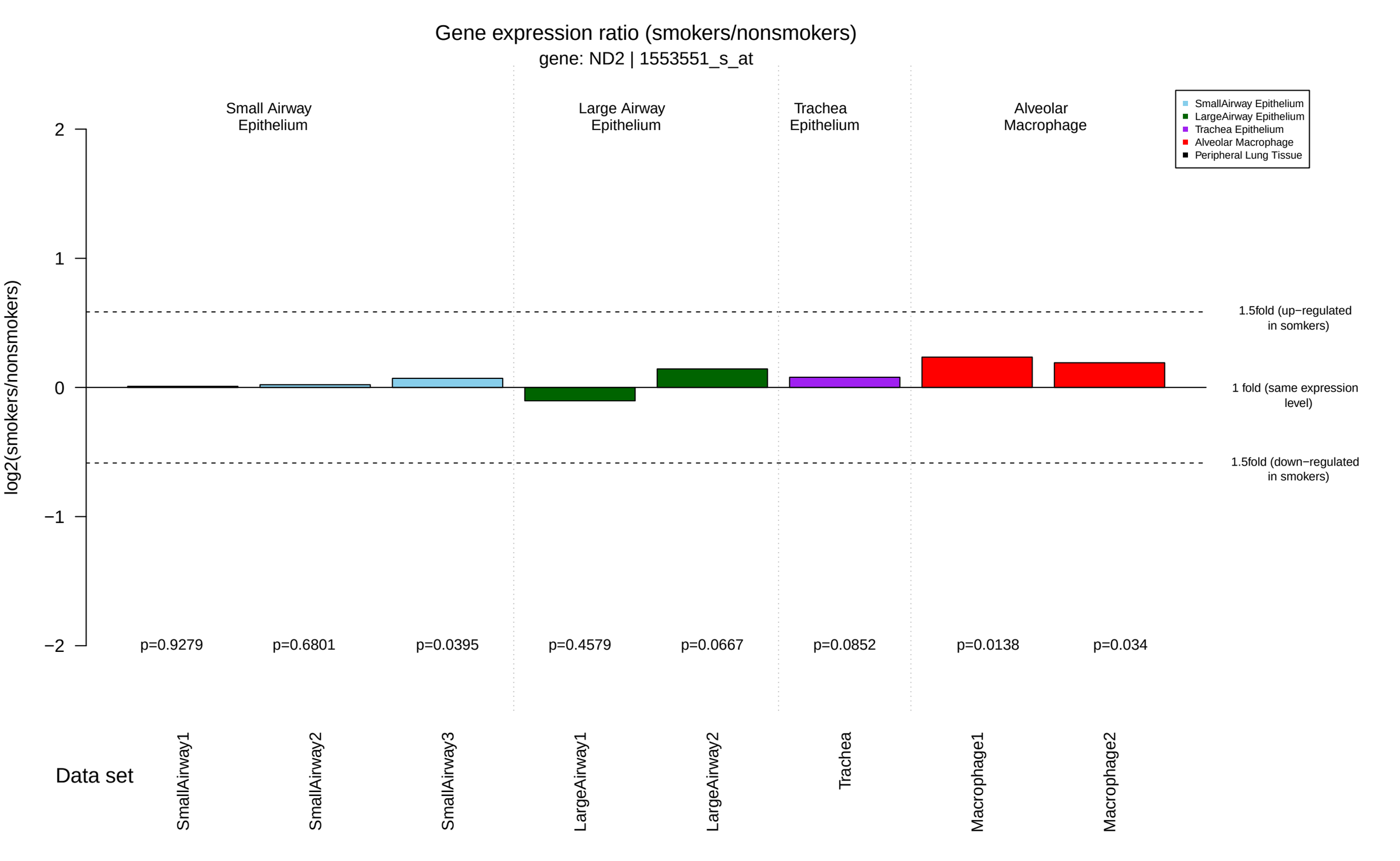
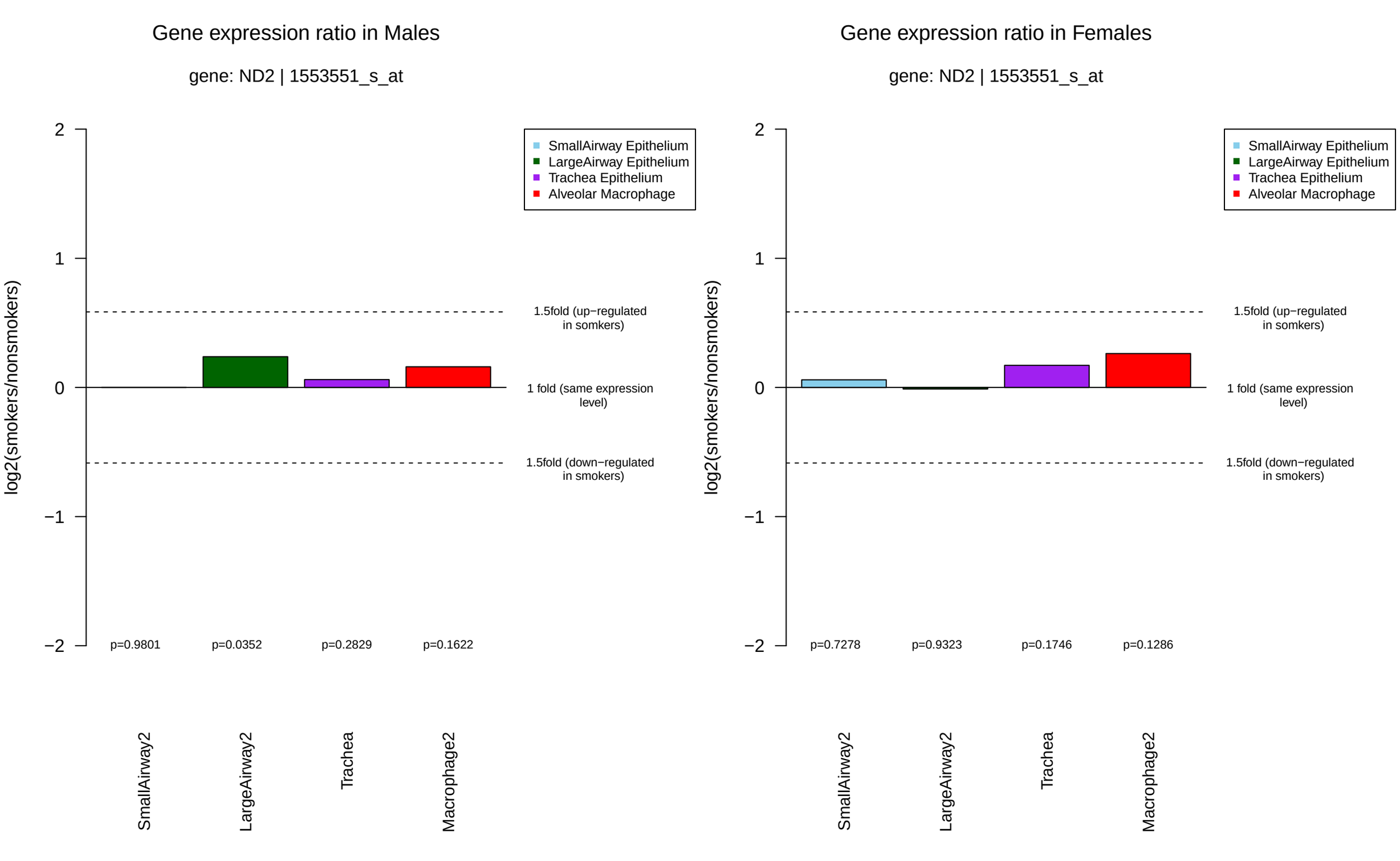
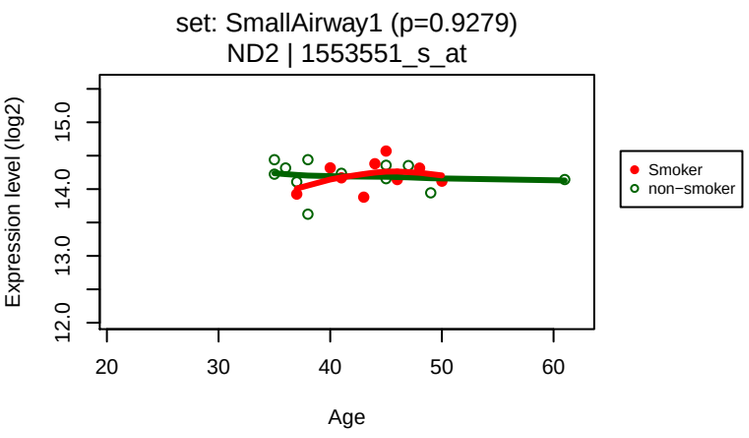
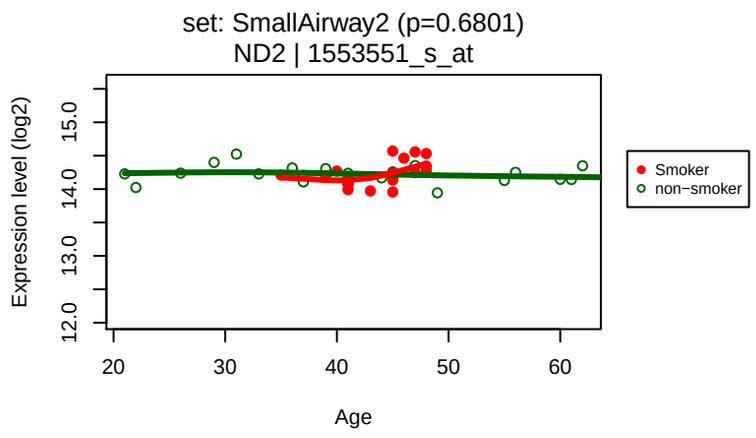
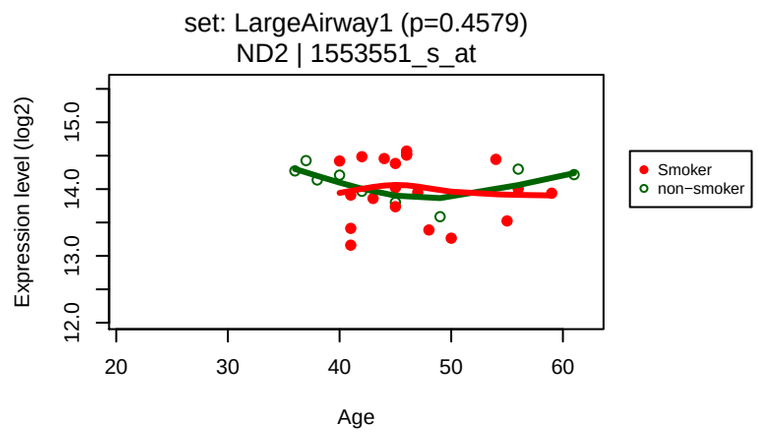

Gene expression of smokers and nonsmokers
gene: ND2 | 1553551_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9279
p=0.6801
p=0.0395
p=0.4579
p=0.0667
p=0.0852
p=0.0138
p=0.034
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

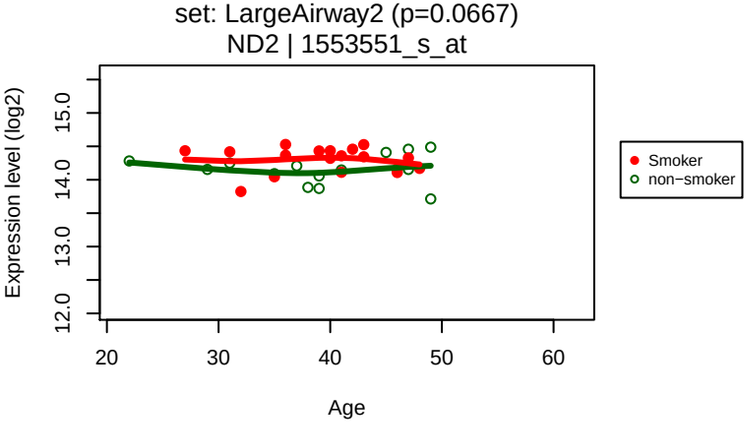

Gene expression ratio (smokers/nonsmokers)
gene: ND2 | 1553551_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9279
p=0.6801
p=0.0395
p=0.4579
p=0.0667
p=0.0852
p=0.0138
p=0.034
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

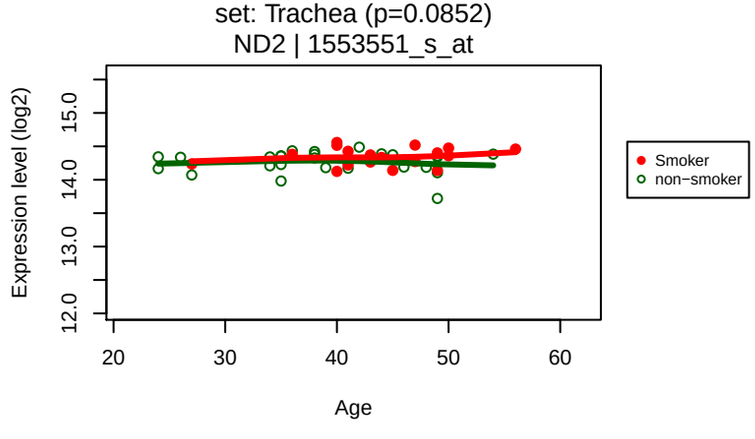

Gene expression ratio in Males
gene: ND2 | 1553551_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9801
p=0.0352
p=0.2829
p=0.1622
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ND2 | 1553551_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7278
p=0.9323
p=0.1746
p=0.1286
SmallAirway2
LargeAirway2
Trachea
Macrophage2

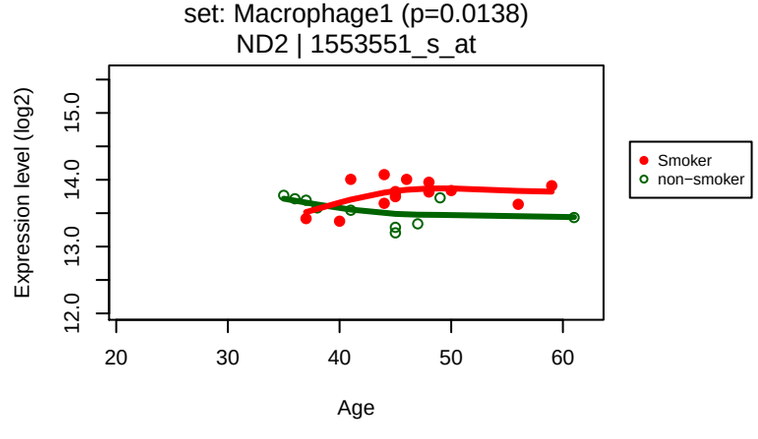

set: SmallAirway1 (p=0.9279)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

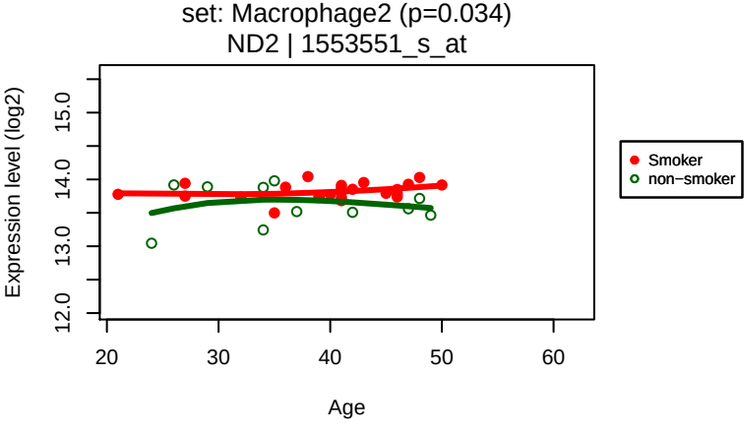

set: SmallAirway2 (p=0.6801)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

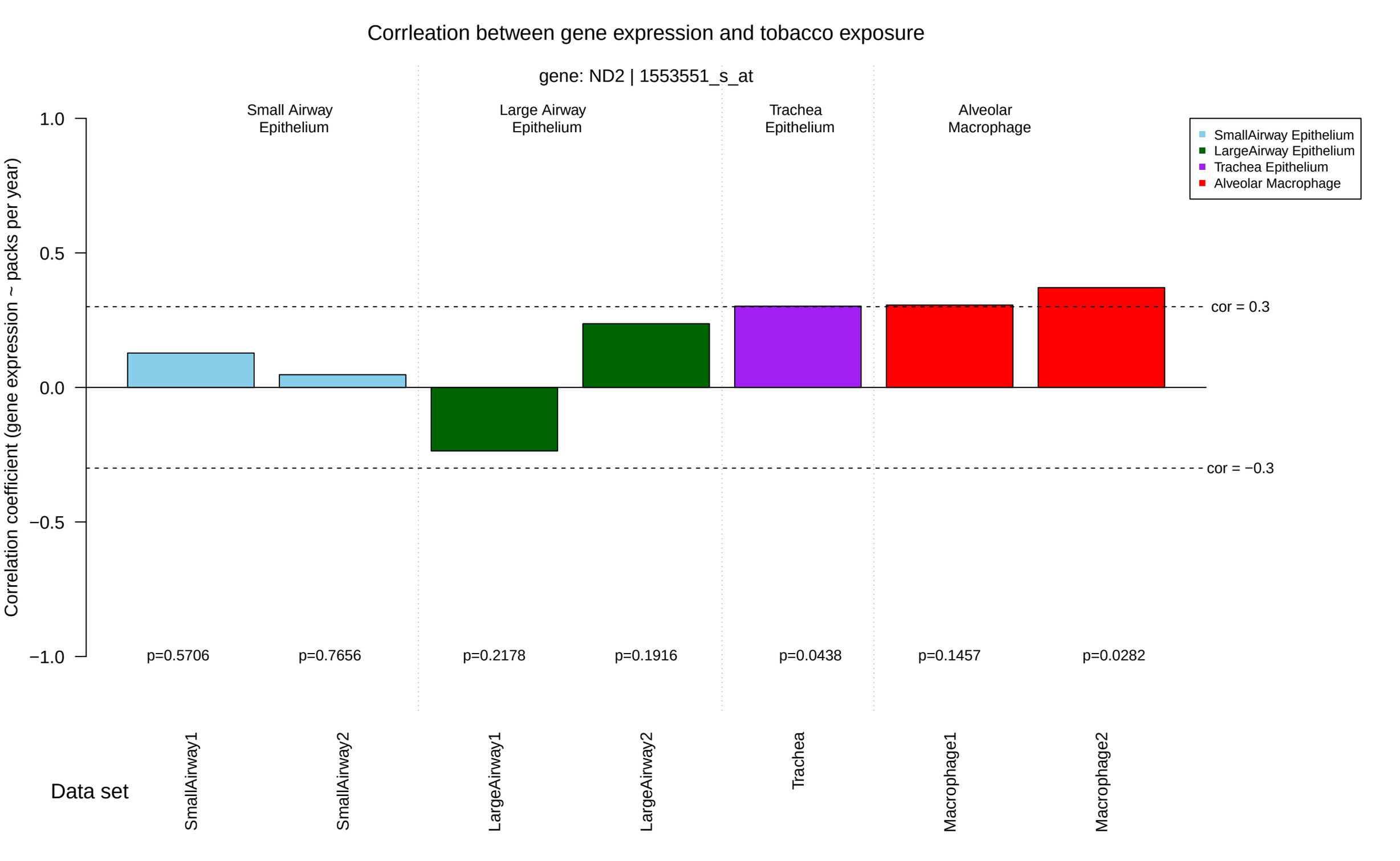

set: LargeAirway1 (p=0.4579)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0667)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0852)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0138)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.034)
ND2 | 1553551_s_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ND2 | 1553551_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5706
p=0.7656
p=0.2178
p=0.1916
p=0.0438
p=0.1457
p=0.0282
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2